Criticality, Boundaries, Self-Organization, and Their Correspondence to the Five-Stage Model in Complexity Science

2026-03-23

Contents

Criticality, Boundaries, Self-Organization, and Their Correspondence to the Five-Stage Model in Complexity Science **2**

1. Purpose and Questions of the Investigation 2
2. Method of Investigation 3
3. Overview of the Model 4
4. Results of the Investigation: Overall Picture 4
5. Results of the Investigation: Examination of Major Theories 5
6. Cross-Cutting Patterns 11
7. Unresolved Questions 12
8. Conclusion 12

Criticality, Boundaries, Self-Organization, and Their Correspondence to the Five-Stage Model in Complexity Science

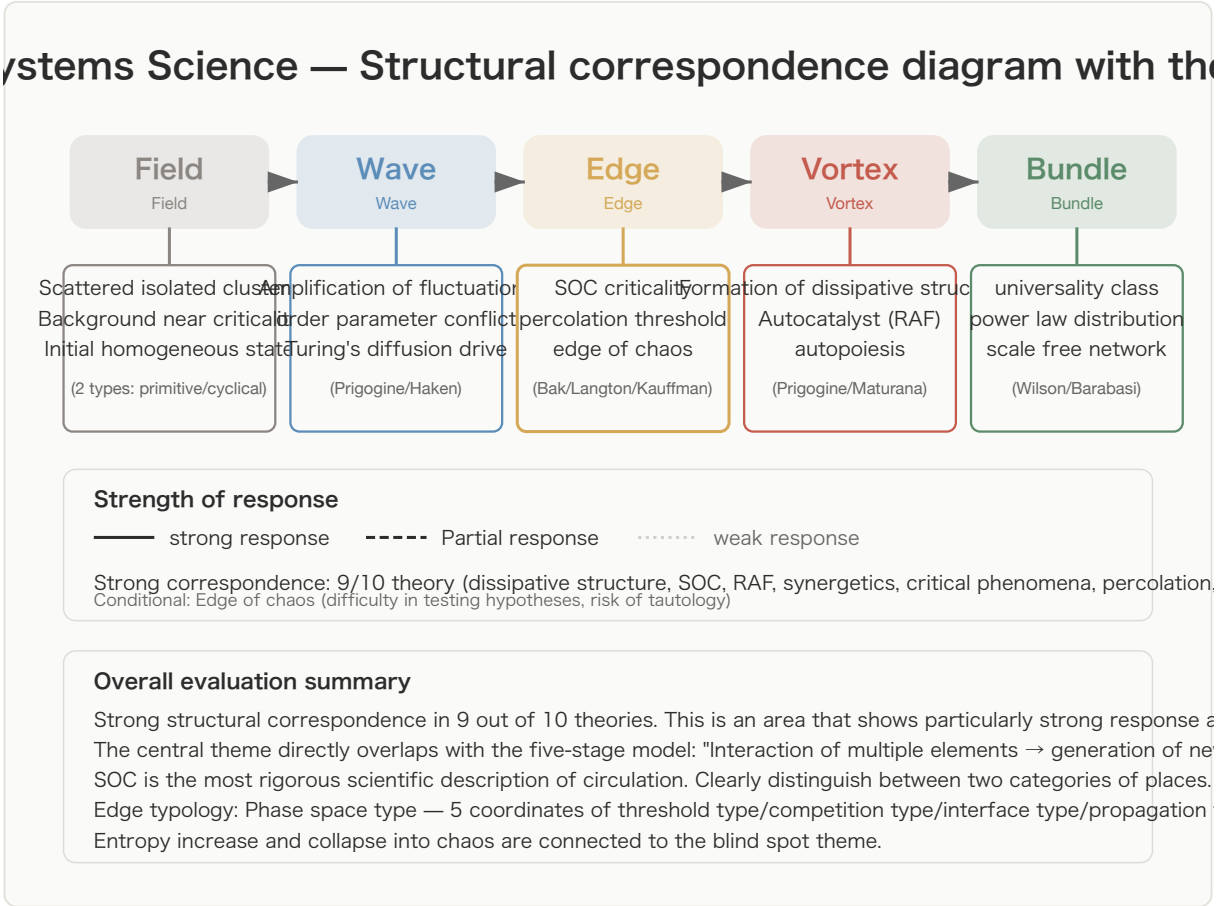

Figure 1: Criticality, Boundaries, Self-Organization, and Their Correspondence to the Five-Stage Model in Complexity Science — Structural correspondence diagram

1. Purpose and Questions of the Investigation

This report was prepared to answer the following question:

> Do the various theories in this academic domain correspond structurally to the five-stage model (Field ( ) -> Wave ( ) -> Edge ( ) -> Vortex ( ) -> Bundle ( ))?

The present domain concerns the structural resemblance between dissipative structures, self-organized criticality, autocatalytic sets, synergetics, critical phenomena, percolation, autopoiesis, reaction-diffusion, the edge of chaos, and the staged generation of order in network science, on the one hand, and the five stages on the other. Across the domain as a whole, structural resemblance is very high. In particular, Edge, understood in terms of criticality, thresholds, boundary conditions, phase transitions, and connective structure, is a central theme throughout complexity science as a whole, and in 8 of the 10 entries it shows especially strong correspondence.

This report presents the results of an investigation; it is not a document written to justify any particular theory. Places where the correspondence is strong and places where reservations are necessary are described with the same degree of neutrality.

Figure 2: Criticality, Boundaries, Self-Organization, and Their Correspondence to the Five-Stage Model in Complexity Science —

2. Method of Investigation

Scope and Selection Criteria

In this domain, 10 theories and practical models were selected for evaluation. The selection was based on three criteria: (1) the theory explicitly treats processes of generation or transformation; (2) structural analysis is possible on the basis of primary sources or well-established secondary sources; and (3) different lineages within the domain can be compared.

Evaluation Method

For each theory, the report first explains what the theory describes and then examines where correspondences with the five stages can be identified. Judgments of correspondence were made by considering together the concepts found in the texts, concrete examples, connections between stages, the presence or absence of a cycle from Bundle to the next Field, and the risk of forced interpretation.

Criteria of Judgment

- **Strong structural correspondence**: The core concepts of the theory correspond concretely to multiple stages of the five-stage model, and the report can explain why this is so in the main text.
- **Conditional correspondence**: A correspondence can be indicated, but interpretive reservations remain in mapping the stages.
- **No correspondence confirmed**: Partial approximations exist, but the theory cannot be read without strain as exhibiting the structure of the five stages as a whole.

Limits of the Investigation

This investigation is not an exhaustive examination of every work associated with each theory. It focuses on the core concepts of major works and extracts structures at a level of granularity that permits comparison. Accordingly, there are places where the internal details of a theory or historiographical disagreements within scholarship could not be fully addressed.

3. Overview of the Model

The five-stage model describes the creative process in terms of the following five stages.

Stage	Definition
Field	An undifferentiated state. It refers to initial conditions in which neither direction nor structure has yet been determined.
Wave	A stage of exploration in which multiple directions diverge and differences are amplified.
Edge	A stage of tension in which opposing elements coexist and do not immediately converge on either side.
Vortex	A stage in which a new coherence or order emerges out of tension.
Bundle	A stage in which form becomes determinate and stabilizes as a reusable structure.

This model is not a one-way line but includes a cycle in which Bundle becomes the condition for the next Field. Moreover, the five stages are not strictly separated boxes but are understood as a continuous spectrum. It includes the hypothesis that the same structure can recur at different scales, from momentary thought to institutional formation to long-term historical change.

If a sense of incongruity remains at this point, that itself is an important concern of the present investigation. Readers may continue with that sense of incongruity still intact.

4. Results of the Investigation: Overall Picture

List of Theories Evaluated

Theory	Proponent	Structural Correspondence
Dissipative structures and order from fluctuations (Prigogine)	Ilya Prigogine (1917-2003). Nobel Prize in Chemistry, 1977	Strong
Self-organized criticality (SOC) and scale-invariant avalanches (Per Bak)	Per Bak (1948-2002), Chao Tang, Kurt Wiesenfeld. BTW sandpile model (1987)	Strong
Autocatalytic sets and the origin of order (Kauffman / Hordijk-Steel)	Stuart Kauffman (1939-). Formalization of RAF theory: Wim Hordijk & Mike Steel (2004-)	Strong

Theory	Proponent	Structural Correspondence
Synergetics and order parameters (Haken)	Hermann Haken (1927-2024). Stuttgart school	Strong
Critical phenomena and universality (Wilson, renormalization group)	Kenneth G. Wilson (1936-2013). Nobel Prize in Physics, 1982	Strong
Percolation and the emergence of macroscopic connectivity	Mathematical formulation: Broadbent & Hammersley (1957). Textbook synthesis: Stauffer & Aharony (1994)	Strong
Autopoiesis and the self-production of boundaries (Maturana & Varela)	Humberto Maturana (1928-2021) & Francisco Varela (1946-2001)	Strong
Reaction-diffusion and the Turing mechanism (Turing)	Alan Turing (1912-1954). 1952, "The Chemical Basis of Morphogenesis"	Strong
The edge of chaos	Christopher Langton (1990). Including Stuart Kauffman's concept of "order for free"	Conditional
Network science and multi-layer / higher-order interactions	Watts & Strogatz (1998, small-world), Barabasi & Albert (1999, scale-free), Battiston et al. (2020, higher-order interactions)	Strong

Of the 10 items evaluated, 9 showed strong structural correspondence, 1 showed conditional correspondence, and 0 failed to show any confirmed correspondence. Overall, this domain exhibits correspondence with the five-stage model at a comparatively high density, but the manner of correspondence is not uniform. Some theories offer a thick description of the conditions of the Field, while others describe with precision the tensions of the Edge or the fixation of the Bundle.

What is especially important is not the high rate of correspondence in itself, but rather which theory describes which stage and in what vocabulary. Even when this domain addresses the same topic of creation, different layers, such as body, institution, time, perception, practice, and symbol, illuminate different moments of the five stages.

5. Results of the Investigation: Examination of Major Theories

5.1 Dissipative structures and order from fluctuations (Prigogine)

In Ilya Prigogine's (1917-2003; Nobel Prize in Chemistry, 1977) theory of dissipative structures and order from fluctuations, open systems under conditions far from equilibrium, that is, under fluxes of energy and matter, undergo amplification of fluctuations, bifurcation, and transition to a new steady structure, namely a dissipative structure. Order does not arise under linear response alone (Prigogine 1977 Nobel Lecture). Bifurcation points occur when control parameters, such as temperature gradients, exceed critical values. The path taken after bifurcation is selected through symmetry breaking.

Dissipative structures physically and chemically realize the movement Field -> Wave -> Edge -> Vortex. Near equilibrium corresponds to Field; the amplification of fluctuations corresponds to

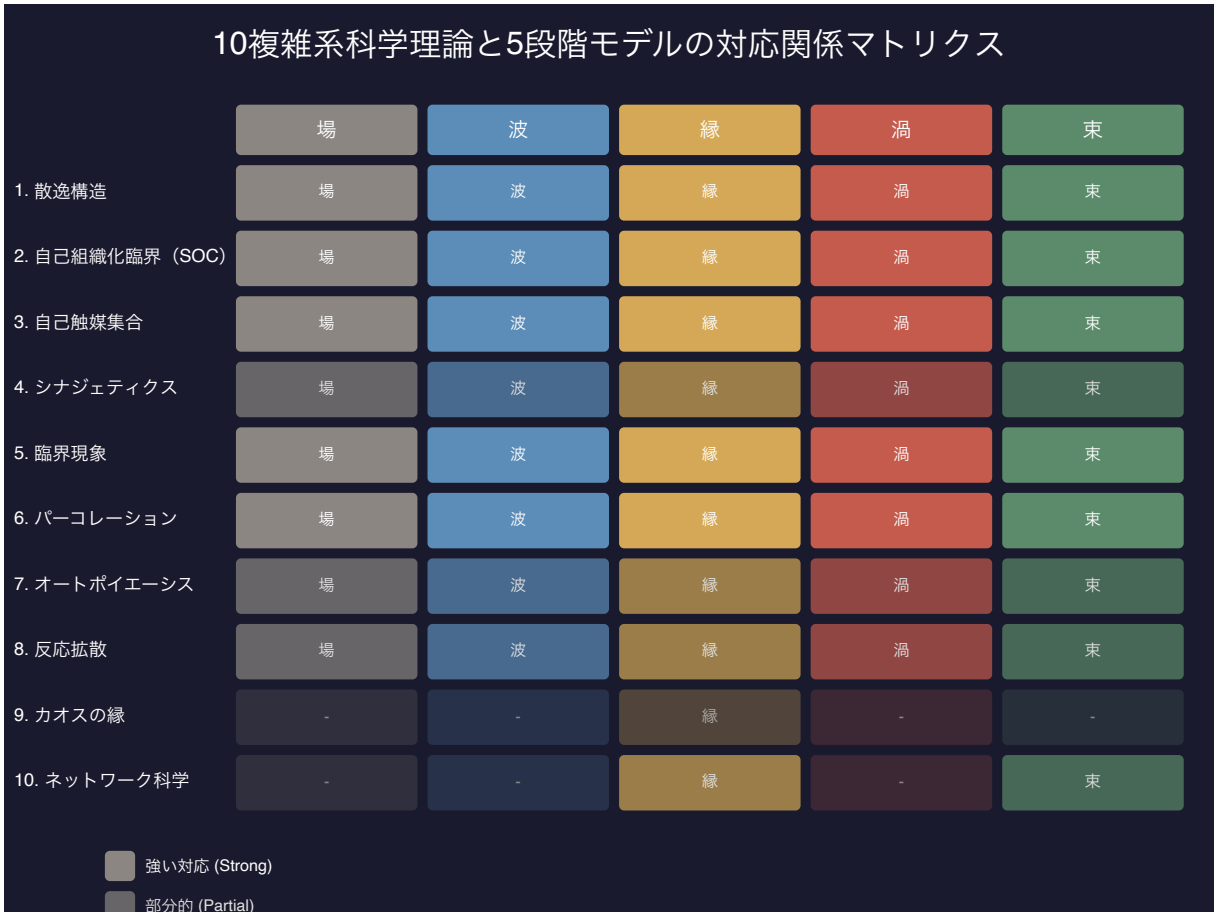

Figure 3: Criticality, Boundaries, Self-Organization, and Their Correspondence to the Five-Stage Model in Complexity Science — $\times 5$

Wave; symmetry breaking at the bifurcation point corresponds to Edge; and the appearance of the dissipative structure corresponds to Vortex. The intensity of the correspondence between the bifurcation point and Edge is especially high, because it arises only in the nonlinear regime and is the site at which a "selection" of path occurs, making it structurally isomorphic with the emergence of relation at a boundary. Benard convection cells offer the clearest visual example: the spontaneous emergence of hexagonal patterns from a uniform temperature gradient makes the passage from Field to Vortex visible.

This work belongs to the context of nonequilibrium thermodynamics. Prigogine's Nobel Lecture (1977) is foundational. It is associated with the Brussels school. Although it developed independently of synergetics (Haken, CX-004), the two overlap as descriptions of nonequilibrium phase transitions. The BZ reaction and Turing patterns in D03 chemistry are concrete examples of dissipative structures.

5.2 Self-organized criticality (SOC) and scale-invariant avalanches (Per Bak)

In Per Bak's (1948-2002), Chao Tang's, and Kurt Wiesenfeld's theory of self-organized criticality (SOC) and scale-invariant avalanches, a system self-organizes into a critical state under slow external driving, and events, namely avalanches, occur on many different scales (Bak, Tang & Wiesenfeld 1987, 1988). Power-law distributions are a principal prediction of SOC, but detecting a power law is itself statistically difficult. Fitting straight lines on log-log plots is inappropriate, and likelihood-based comparison must be combined with goodness-of-fit testing (Clauset et al. 2009).

SOC corresponds strongly to all five stages: background near criticality (Field) -> local threshold crossing (Wave) -> neighborhood interaction (Edge) -> avalanche (Vortex) -> power law (Bundle). In addition to this correspondence, a decisive point is that the cycle from Bundle back to Field is naturally internal to the theory itself. After an avalanche, the system is again slowly driven and moves toward the next critical state. The cyclicality of the five stages is thus described by the theory itself.

Scale invariance, in the form of the power law, corresponds clearly to Bundle. The statistical structure of the event population quantitatively captures the accumulation of generation. This differs qualitatively from the "maintenance-structure" type of Bundle seen in dissipative structures (CX-001). In SOC, the power law is the statistical trace left by many generative events and is the purest realization of a "statistical-structure" type of Bundle. At the same time, careless assertions of power laws collapse into a condition in which everything can be explained and therefore nothing is explained. Clauset et al. (2009) should be treated as the reference standard, and claims about power laws should always be accompanied by this validation procedure. This work belongs to the context of statistical physics and complexity science. Bak (1987/1988) is foundational. It also connects with the edge of chaos (Langton 1990, CX-009): SOC describes the dynamics by which a system spontaneously approaches criticality, whereas the edge of chaos asks about the functional significance of being near criticality as such. Its relation to critical phenomena (CX-005) is that SOC concerns the dynamics of reaching criticality, whereas critical phenomena formalize what criticality is.

5.3 Autocatalytic sets and the origin of order (Kauffman / Hordijk-Steel)

In Stuart Kauffman's (1939-) theory of autocatalytic sets and the origin of order, together with the formalization of RAF theory by Wim Hordijk and Mike Steel (2004-), chemical reaction networks can form self-catalytically closed sets, that is, RAFs (Reflexively Autocatalytic and Food-generated sets), under certain conditions, specifically when catalytic density parameters exceed threshold values (Kauffman 1986, 1993). RAF theory (Hordijk & Steel 2004) provided an efficient algorithmic formalization for detecting autocatalytic sets.

Autocatalytic sets describe the transition from Edge to Vortex as the formation of closure. Exceeding the threshold of catalytic density parameters, that is, phase-transition-like emergence, corresponds to Wave -> Edge, while the interface between the food set and the RAF is itself the structure of Edge, because the theory itself defines what counts as inside and what counts as outside. The formation of closure, namely the emergence of a self-maintaining bundle of reactions, corresponds to Vortex, and this is a chemical realization of individuation.

This work belongs to the context of origin-of-life research and complex systems biochemistry. Kauffman (1986, 1993) is foundational, and RAF theory (Hordijk & Steel) supplies the mathematical formalization. Connections with recent RNA experimental systems are advancing. It resonates deeply with autopoiesis (CX-007) around the question of system autonomy.

5.4 Synergetics and order parameters (Haken)

In Hermann Haken's (1927-2024; Stuttgart school) synergetics and order-parameter theory, nonequilibrium phase transitions in systems with many degrees of freedom are described in a unified way through order parameters and the slaving principle. Unstable modes "enslave" stable modes, and degrees of freedom are dramatically reduced (Haken 1977, 1983). The laser is the paradigmatic case: the radiation of many atoms becomes spontaneously cooperative, generating coherent light.

Synergetics describes the transition from Wave to Vortex through the slaving principle. The process by which unstable modes, that is, dominant fluctuations within Wave, pass through competition among modes at Edge and become established as an order parameter at Vortex is structurally isomorphic with the movement from fluctuation to relation at the boundary to the emergence of a new coherence in the five-stage model. Laser oscillation is the clearest physical realization of this process: the disordered radiation of many atoms spontaneously converts into coherent emission.

This work belongs to the context of nonequilibrium physics and cooperative phenomena. Haken (1977) is foundational. It is associated with the Stuttgart school. It developed in parallel with, and independently from, the theory of dissipative structures associated with the Brussels school. Its applications have spread from laser physics into fluid dynamics, neuroscience, and pattern formation.

5.5 Critical phenomena and universality (Wilson, renormalization group)

In Kenneth G. Wilson's (1936-2013; Nobel Prize in Physics, 1982) theory of critical phenomena and universality, the correlation length diverges at the critical point, and the entire system becomes scale-invariant. Universality means that there are systems with different microscopic details that nevertheless share the same critical exponents (Wilson 1971, 1983 Nobel Lecture). The renormalization group (RG) is a mathematical method for tracing the behavior of systems under scale transformation. Flow toward fixed points characterizes critical phenomena.

Critical phenomena correspond very strongly to the five stages. In particular, the correspondences Edge = critical point, where couplings extend across all scales, and Bundle = universality class, that is, structure beyond microscopic detail, are powerful. Divergence of correlation length (Wave) -> critical point (Edge) -> new phase (Vortex) -> universality (Bundle) can be regarded as the mathematical skeleton of the five-stage model. The singularity of the critical point, as the only place where all scales are coupled and microscopic details can determine macroscopic behavior, is the most rigorous physical description of Edge.

The concept of universality provides an insight unavailable in any other entry. The question of why different systems display the same behavior is structurally isomorphic with the fundamental

question of this investigation: why the same five-stage structure appears across 30 different academic domains. What the renormalization group teaches is the mechanism by which microscopic detail disappears at the macroscopic level. This remains a suggestion at the level of implication, but it offers a candidate physical answer to the question of why structural resemblance is possible. The realization of Bundle as universality is a third type distinct from the maintenance-structure type in dissipative structures and the statistical-structure type in SOC: a universality type, in which different microscopic systems share the same macroscopic structure. This work belongs to the context of statistical physics and phase transitions. Wilson (1971) is foundational. It marks the development from Landau theory to the renormalization group. It is also the physical foundation of SOC (Bak, CX-002), and percolation (CX-006) is likewise positioned as one field within critical phenomena. It connects directly with the phase transition entry in D02 physics.

5.6 Percolation and the emergence of macroscopic connectivity

In percolation and the emergence of macroscopic connectivity, mathematically formulated by Broadbent & Hammersley (1957) and synthesized in textbook form by Stauffer & Aharony (1994), when the occupation probability p of sites or bonds on a lattice or network exceeds the threshold p_c, a macroscopically connected component, namely an infinite cluster or spanning cluster, appears (Broadbent & Hammersley 1957). Near p_c, the cluster-size distribution becomes a power law, making percolation one kind of critical phenomenon. Critical exponents depend on the dimension and type of lattice, but are classified into universality classes (Stauffer & Aharony 1994).

Percolation describes the movement Field -> Wave -> Edge -> Vortex in terms of the minimal element of whether things connect or not. The percolation threshold p_c is the purest mathematical realization of Edge: the accumulation of local connections gives rise, at a single point, to a macroscopically distinct structure, namely a giant cluster. As a point of qualitative transformation, p_c lays bare the percolation-like structure inherent in the bifurcation point of dissipative structures, the avalanche chains of SOC, and the formation of closure in autocatalytic sets.

This work belongs to the context of probability theory and statistical physics. Broadbent & Hammersley (1957) mark its origin. Although it is positioned as one field within critical phenomena (CX-005), its connection with network science (CX-010) is also strong. It connects directly with D01 mathematics (bifurcation theory) and D02 physics (phase transitions).

5.7 Autopoiesis and the self-production of boundaries (Maturana & Varela)

In Humberto Maturana's (1928-2021) and Francisco Varela's (1946-2001) theory of autopoiesis and the self-production of boundaries, an autopoietic system is a circular organization in which components produce the network that produces them, and the network in turn produces the components (Maturana & Varela 1980). The self-production of the boundary, that is, the membrane, is the core of the theory: the system maintains its self-identity by continuously producing its own boundary (Varela, Maturana & Uribe 1974).

Autopoiesis places at the center of the theory a cycle in which Edge creates Vortex and Vortex creates Edge. The self-production of the boundary, namely the membrane (Edge), leads to the maintenance of self-identity (Vortex), and that Vortex in turn reproduces the boundary (Edge). This circular structure gives the most explicit description of the transition Edge <-> Vortex in the five-stage model. In order to stand as an individual, that is, as Vortex, a system must continue to produce its own boundary, that is, maintain Edge. The two are inseparable.

This work belongs to the context of theoretical biology and cognitive science. Maturana & Varela (1980) are foundational. It is associated with the Chilean school. There are extensions into

sociology (Luhmann), developments in cognitive science (enactivism; Varela/Thompson/Rosch), and connections with artificial life research. It resonates deeply with autocatalytic sets (CX-003) around the question of system autonomy. Its connection with D13 philosophy, especially Simondon's theory of individuation, is also important.

5.8 Reaction-diffusion and the Turing mechanism (Turing)

In Alan Turing's (1912-1954) reaction-diffusion and Turing mechanism, presented in the 1952 paper "The Chemical Basis of Morphogenesis," a homogeneous steady state is destabilized by diffusion-driven instability, and spatial patterns form spontaneously (Turing 1952). The mechanism consists in the interaction of an activator, which acts over short range and self-amplifies, and an inhibitor, which acts over long range and suppresses. Differences in diffusion speed generate the pattern.

Reaction-diffusion provides a correspondence to Field -> Wave -> Vortex through the movement from homogeneous state to destabilization to spatial pattern. The mechanism by which a difference in diffusion speed between activator and inhibitor generates pattern is a concrete realization of the process by which amplification of difference (Wave) crystallizes into spatial structure through relation (Edge) and gives rise to Vortex. If dissipative structures (CX-001) provide the general framework, reaction-diffusion provides the concrete mechanism of spatial pattern formation.

This work belongs to the context of mathematical biology and morphogenesis. Turing (1952) is foundational. It can be positioned as a concrete mechanism within dissipative structures (CX-001), but it also has its own distinct reach through pattern selection by boundary conditions. It overlaps directly with the BZ reaction and Turing patterns in D03 chemistry: D03 treats them as chemical reactions, whereas D29 treats them as mechanisms of pattern formation in complexity science.

5.9 The edge of chaos

In Christopher Langton's (1990) edge of chaos, including Stuart Kauffman's concept of "order for free," there is a possibility that information-processing and computational capacity are maximized in the parameter space of cellular automata near the phase transition between the ordered phase and the chaotic phase (Langton 1990). In Kauffman's random Boolean networks (RBNs), order appears spontaneously near $K = 2$, at the boundary between freezing and chaos.

The edge of chaos directly asserts the functional significance of remaining at the boundary. By being near the phase transition, neither in order, that is, freezing, nor in chaos, that is, runaway dynamics, computational and information-processing capacities are maximized. This is the most direct scientific expression of S57, "creation arises through remaining at the edge (boundary)." Whereas other entries describe passing through Edge, the edge of chaos argues for the value of remaining at Edge.

In the other entries, Edge appears as a point or moment to be passed through, whereas in the edge of chaos it is positioned as a regime in which one remains. This emphasis on remaining has the most direct structural connection with the function of maintaining tension in D3: rather than reflexively moving toward order, that is, resolution, or toward chaos, that is, abandonment, remaining at the edge is itself creative. At the same time, the appeal of this concept is also its risk. As Mitchell et al. (1993) pointed out, the precise characterization of the edge of chaos is difficult, and the concept has circulated widely without a stable operational definition. This work belongs to the context of artificial life, cellular automaton theory, and complex adaptive systems. Langton (1990) is foundational. It connects with SOC (CX-002): SOC describes the dynamics by which a system spontaneously approaches criticality, whereas the edge of chaos asks

about the functional significance of being near criticality as such. Kauffman's "order for free" stands at the intersection of SOC and the edge of chaos.

5.10 Network science and multi-layer / higher-order interactions

In network science and multi-layer / higher-order interactions, as developed by Watts & Strogatz (1998, small-world), Barabasi & Albert (1999, scale-free), and Battiston et al. (2020, higher-order interactions), small-world structure means that a high clustering coefficient and a short average path length coexist at an intermediate point between random graphs and regular lattices (Watts & Strogatz 1998). Scale-free structure means that preferential attachment gives rise to networks whose degree distributions follow a power law (Barabasi & Albert 1999). At the same time, Broido & Clauset (2019) showed empirically that scale-free networks are rare.

Network science takes as its central thesis that the structure of interactions determines dynamics. This is a formalization of the five-stage claim that Edge, understood as the structure of connection and relation, determines Vortex, understood as emergent pattern. Small-world structure, where a small number of bridge connections link distant regions, describes the efficiency of Edge, while scale-free structure, where a small number of hubs connect many nodes, describes the heterogeneity of Edge. The fact that percolation (CX-006) also holds on networks, for example as threshold behavior in robustness under node removal, demonstrates the consistency of theories of Edge across the domain.

Whereas the other entries ask what happens at Edge, such as selection at bifurcation points, transformation at thresholds, and boundary production at interfaces, network science asks what kind of structure Edge has. It is therefore a structural theory of Edge, complementary to the process theories of Edge represented by the other entries. Higher-order interactions (Battiston et al. 2020) are especially important, because they formalize collective interactions involving three or more nodes, rather than only pairwise interactions, by means of hypergraphs and simplicial complexes. This provides theoretical support for the possibility that condition (1) of Edge's three conditions, namely the relational network, need not be reducible to simple pairs but may instead be genuinely many-body. This work belongs to the trajectory from graph theory to social network analysis to the science of complex networks. It developed explosively through the data-driven approaches of the late 1990s, especially Watts-Strogatz and Barabasi-Albert. It is closely connected with percolation (CX-006) and critical phenomena (CX-005). Network percolation, that is, the threshold disappearance of the giant connected component under node removal, is foundational for network robustness.

6. Cross-Cutting Patterns

First, as shown by dissipative structures and order from fluctuations (Prigogine), self-organized criticality (SOC) and scale-invariant avalanches (Per Bak), and autocatalytic sets and the origin of order (Kauffman / Hordijk-Steel), this domain describes difference and opposition not merely as obstacles but as conditions for the rise of a new order. This means that, among the five stages, the transition from Edge to Vortex can be observed here with particular richness.

Second, what synergetics and order parameters (Haken), critical phenomena and universality (Wilson, renormalization group), and percolation and the emergence of macroscopic connectivity share is that generation is not a one-off event but includes a cycle of forming preconditions, amplifying tension, producing local order, stabilizing it, and moving into the next reconfiguration. This confirms that the five-stage model must be read not as a static table of classifications but as a temporal movement.

By contrast, the edge of chaos also shows that correspondence with the five stages is not established automatically. The same number of stages or a resemblance in vocabulary is not

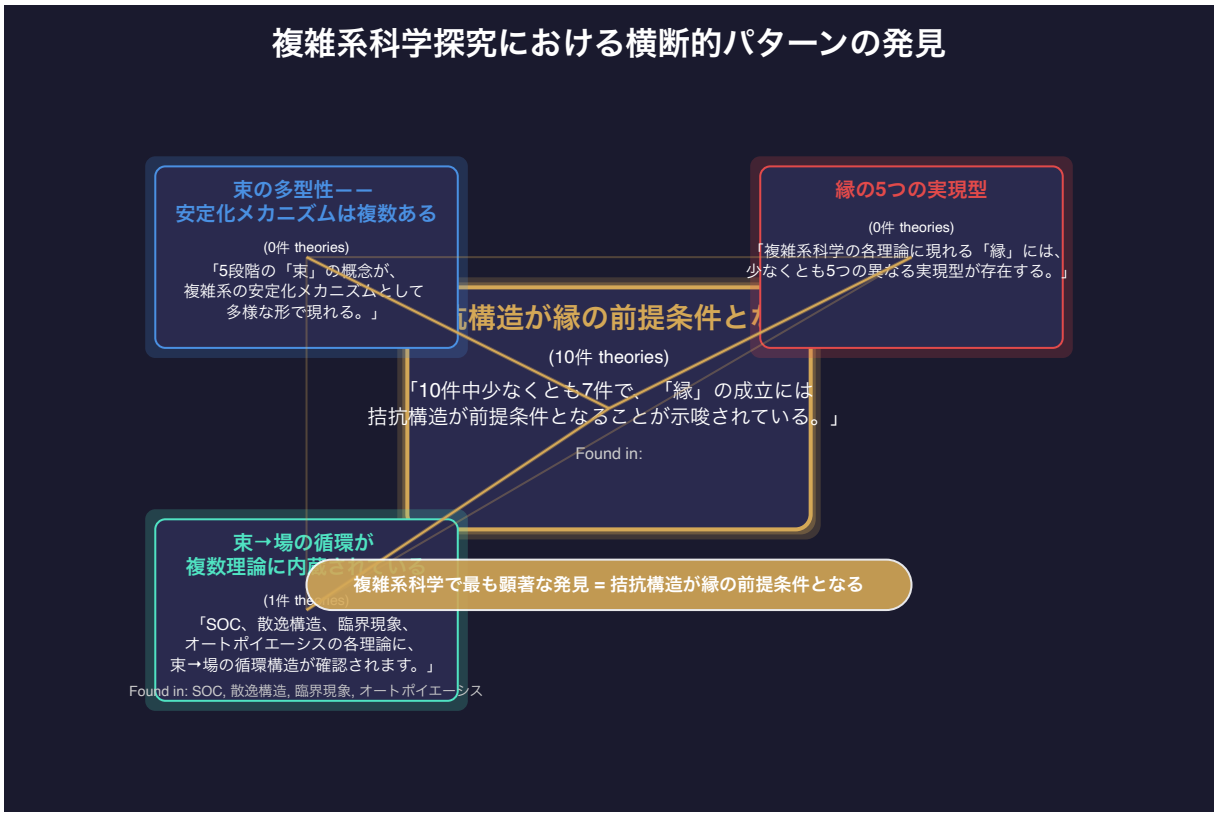

Figure 4: Criticality, Boundaries, Self-Organization, and Their Correspondence to the Five-Stage Model in Complexity Science —

sufficient. It is necessary to examine case by case which concepts bear the points of transformation and at what level the structure emerges.

7. Unresolved Questions

1. To what extent can the edge of chaos be read as a stage description of the five-stage model?
2. To what extent can the structures confirmed in this domain be extended from the explanation of individual cases into a more general model of generation? The conditions under which comparability can be preserved without losing the specificity of the objects still require closer examination.
3. Should the five stages be read as a linear progression, or rather as a cycle including multiple temporal layers and repetitions? In particular, the theories differ in how explicitly they describe the return from Bundle to the next Field.
4. What becomes visible, and what is lost, when the vocabulary specific to this domain is replaced by the general vocabulary of the five stages? The boundary between what is translatable and what is not is itself an important issue for future comparative investigation.

8. Conclusion

This reconsideration of the domain confirmed strong structural correspondence in 9 of the 10 theories and conditional correspondence in 1. In particular, dissipative structures and order from fluctuations (Prigogine), self-organized criticality (SOC) and scale-invariant avalanches (Per Bak), and autocatalytic sets and the origin of order (Kauffman / Hordijk-Steel) provide

especially rich descriptions of the movement from Field to Edge, or from Edge to Vortex, and therefore occupy a central place as evidence within this domain.

At the same time, the findings of this domain do not support the claim that the same five stages simply appear unchanged in every theory. The stages that appear most strongly differ from one theory to another, and in some cases reservations are necessary in mapping the stages. What matters, therefore, is not to generalize directly from a high rate of correspondence, but to continue comparing which moments are described in which vocabulary.

In that sense, this domain does not merely support the validity of the five-stage model in a single direction; it is also a domain that imposes conditions on how the model should be read. The very coexistence of theories that correspond densely and theories that require reservation constitutes the value of the comparative investigation.